DILLON GREEN

FLIGHT SIMULATOR ENGINEER · SOFTWARE & AUTOMATION

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SUMMARY

Flight simulator engineer and self-taught software developer with 10 years in aviation, 8 of them hands-on with Boeing and Airbus avionics, full flight simulators and FAA-qualified training devices. Designed, built and deployed PAR, the ~72k-line parts, procurement and work-order platform used across the 40-person simulator department, and leads the intern-built MPIC test bench. Pairs hands-on simulator expertise with AI-assisted software development to modernize how operations teams work.

SOFTWARE PROJECTS

PAR (Parts & Requisitions Platform) — *Simulator Operations (sole developer)*

Python · Flask · SQLite · Bootstrap · GDAL · PyInstaller

- Designed and built the department's inventory, procurement, work-order and budget system: ~72k lines, 231 routes, 50+ tables, three user roles and a full audit ledger; rolled out to a 40-person department, in daily use.
- Migrated off legacy MS Access and GigaTrak with 15 importers, including a reviewable two-way reconcile engine (field-level diffs, learned vocabulary mapping, idempotent history import that keeps the original timestamps).
- Built a schematics pipeline that converts 6,000+ CAE MicroStation DGN drawings to searchable SVG via GDAL and links part numbers on drawings to live stock.
- Added fleet-level reorder logic across 26 simulator devices at two sites: spares aggregation, critical and low-stock tiers, ABC/FSN classification; plus a REST API, QR labels printed to Zebra printers, and mobile photo capture.
- Backed it with 340+ automated checks (72 regression suites, 142 headless-browser end-to-end checks, a 30-thread concurrency test), a ratcheting quality gate, automated offsite backups and a one-file Windows service deploy.

Linework — *Open-source CAD viewer and editor in the browser (linework.dillongreen.dev)*

JavaScript · Canvas · Web Workers · WebAssembly · pdf.js · Offline PWA · Playwright

- Opens DXF, DWG, MicroStation DGN and PDF drawings entirely on the user's device; measure, mark up, edit and save back to DXF or DGN.
- Traces a wire's whole net on a schematic in under 50 ms on a 53k-entity drawing; indexes a folder of sheets in the background, links off-sheet references and searches text across every sheet.
- Edits DGN files losslessly by patching the original bytes in place; reads DWG with LibreDWG compiled to WebAssembly in a bit-for-bit reproducible build and loaded only on demand, keeping app startup under a 150 KB budget; 330+ automated unit and end-to-end tests.

Cookfold — *Recipe and meal-planning app for iPhone (live on the App Store; formerly Crumb)*

JavaScript · Capacitor · Swift · RevenueCat · Cloudflare · Codemagic CI

- Wrote a native Swift plugin for on-device Vision OCR, iCloud sync and a share extension; local-first sync design; subscriptions; shipped through App Store review with Codemagic CI builds to TestFlight; 460+ automated checks.

pocket429 — *ARINC 429 decoder for avionics technicians (iOS app in App Store review; pocket429.dillongreen.dev)*

JavaScript · Capacitor · Cloudflare Workers · Stripe

- Decodes and builds ARINC 429 data words (BNR, BCD, ISO-5, parity, bit-order detection) against a label database compiled from cited OEM manuals.

Smart Mirror — *Raspberry Pi magic mirror (hardware build in progress)*

Python · Raspberry Pi · systemd · Chromium kiosk · PIR sensor

- Zero-dependency Python server and browser UI behind a two-way mirror: self-starting kiosk, cached proxying that never blanks on network loss, Wi-Fi presence detection, motion-sensor screen wake and phone-to-mirror notes over QR.

Dragon Realm Online — *Real-time multiplayer browser game (play)*

JavaScript · Canvas · Cloudflare Durable Objects · WebSockets · D1

- Live position sync, chat, trading and duels over hibernating WebSockets, with a player market stored in D1 (~11k lines).

Little Airfield — *Cozy airport-builder game for iPhone (playable prototype)*
JavaScript · Canvas (isometric, art drawn in code) · Playwright

- Clear planes to land into the wind, fix pilot-reported squawks and grow a grass strip into a towered airport; pure rules engine with automated tests and iPhone-landscape end-to-end checks.

Also built: Mixtape Drift (rhythm street racer with a WebAudio synth band), surprisepass (live, Stripe payments), halfsies (receipt-OCR bill splitter), and butter and beepbeach (canvas games). Portfolio: dillongreen.dev.

EXPERIENCE

Alaska Airlines Seattle, WA

Flight Simulator Engineer **June 2025 – Present**

- Engineer for Boeing 737, 787 and Airbus A330 full flight simulators on CAE XR, R4, R5 and L3Harris platforms, holding each device to FAA qualification standards.
- Lead the summer intern team building a standalone MPIC test bench: 16 MPICs in a dedicated I/O chassis, driven by a repurposed Windows Server 2019 sim node on its own 48-port network. The bench tests and calibrates I/O and checks aircraft panels from CAE's OneUI without taking a training device offline; loading the aircraft simulation onto it is the next phase.
- Write code-level fixes across simulator platforms for faults that hardware or configuration changes can't resolve.
- Design and implement MPIC interface modifications that improve system reliability.
- Built and deployed PAR, the department's parts, procurement and work-order platform (see Software Projects).
- Work with regulatory and pilot-training teams to sustain FAA-mandated qualification.

Flight Simulator Technician III **2022 – June 2025**

- Maintained and troubleshot CAE and L3Harris full flight simulators, clearing time-critical faults before scheduled training sessions.
- Integrated new hardware through MPIC interface management; subject-matter expert for ISS/IOS issues.
- Resolved recurring discrepancies from schematics, wiring diagrams and pilot handbooks; kept FAA qualification records.

Panasonic Avionics — *Maintenance Services Technician* 2018 – 2022

- Performed systems verification and in-flight entertainment software validation inside Boeing 737 MAX and 787 production facilities ahead of customer delivery.
- Worked on aircraft at Seattle-Tacoma International Airport (SEA), diagnosing complex faults across Boeing 737, 757, 767, 787 and Airbus A320 and A330 fleets and running software installs and end-to-end system checkouts in live airline operations.
- Supported engineers on electrical troubleshooting using wiring diagrams and system schematics.

SkyWest Airlines — *Parts Clerk I (2018); Cross-Utilized Agent (2016–2018)* 2016 – 2018

- Managed parts inventory and sourcing for CRJ and ERJ line maintenance; earlier, ramp, gate and customer-service operations.

SKILLS

- **AI & tooling:** AI-assisted development and code review (Claude Code, Anthropic API)
- **Languages:** Python, JavaScript, Swift, SQL, PowerShell; working knowledge of C++, C#, Fortran
- **Platforms:** Flask, SQLite/PostgreSQL, Cloudflare Workers, Capacitor (iOS & Android), RevenueCat, GDAL
- **Engineering practice:** automated regression and headless-browser testing, CI (Codemagic), data migration and reconciliation, backup/restore, Windows service deployment
- **Aviation:** Boeing 737/787 and Airbus A320/A330 systems, CAE XR/R4/R5 and L3Harris simulators, MPIC and ISS/IOS, ARINC 429, electrical schematics, FAA simulator qualification
- **Certification:** CAE XR Maintenance Certificate